The One-Step Subspace in the Quantum Wielandt Inequality

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1 The assertion

The quantum Wielandt inequality of Sanz–Pérez-García–Wolf–Cirac is stated in [SPGWC10, Theorem 1].[1] Let $\mathcal{E}_A$ be a quantum channel on $M_D(\mathbb{C})$ with Kraus operators $A_1, \dots, A_a$. For each n, set

$$S_n(A) = \operatorname{span}\{A_{i_1} \cdots A_{i_n} : 1 \le i_1, \dots, i_n \le a\}.$$

Thus $S_n(A)$ is the subspace generated by words of length exactly n. Let

$$i(A) = \min\{n : S_n(A) = M_D(\mathbb{C})\},$$

when this minimum exists.

The paper writes d for the number of linearly independent Kraus operators. Equivalently, after replacing the displayed Kraus list by a minimal spanning list, one has

$$d = \dim S_1(A).$$

With this convention, Theorem 1 asserts that a primitive channel satisfies

$$i(A) \le (D^2 - d + 1)D^2.$$

The same theorem also gives two better estimates under additional hypotheses on the one-step subspace. If $S_1(A)$ contains an invertible matrix, then

$$i(A) \le D^2 - d + 1.$$

[1] For traceability, this note corresponds to issue #1049. The local source passage is `Papers/0909.5347/main.tex:360--365`, `Papers/0909.5347/main.tex:572--590`, and `Papers/0909.5347/main.tex:645--684`.

If $S_1(A)$ contains a non-invertible matrix with a nonzero eigenvalue, then

$$i(A) \le D^2.$$

These are hypotheses about arbitrary matrices in $S_1(A)$. The paper does not require the special matrix to be one of the Kraus operators in a chosen Kraus representation.

2 What has been formalized

The formal statements use the same subspaces $S_n(A)$ and define the Kraus rank by

$$\mathrm{kr}(A) = \dim S_1(A).$$

This is the paper's number d written invariantly. Replacing d by $\mathrm{kr}(A)$ is therefore not a change in the mathematical bound. [2]

The general inequality is formalized for the same index $i(A)$:

$$i(A) \le (D^2 - \mathrm{kr}(A) + 1)D^2.$$

The proof passes through the usual eventual full-rank formulation and a blocking argument, but the conclusion is the equality $S_n(A) = M_D(\mathbb{C})$ for one fixed length n. It is not merely a statement about the increasing sum $S_0(A) + \cdots + S_n(A)$.[3]

The two improved estimates now use the paper's one-step subspace hypotheses. For the invertible case, the formal statement assumes a matrix $X \in S_1(A)$ and X invertible; under that assumption it proves

$$S_{D^2-\mathrm{kr}(A)+1}(A) = M_D(\mathbb{C}).$$

For the non-invertible case, the formal statement assumes a matrix $X \in S_1(A)$, X non-invertible, and a nonzero eigenvalue of X; under that assumption it proves

$$S_{D^2}(A) = M_D(\mathbb{C}).$$

[2] The formal definitions are MPSTensor.wordSpan in `TNLean/Wielandt/SpanGrowth/CumulativeSpan.lean:57--64` and MPSTensor.krausRank in `TNLean/Wielandt/Primitivity/PaperDefinitions.lean:85--91`. The local source identifies d as the number of linearly independent Kraus operators in `Papers/0909.5347/main.tex:360--365`; the proof of Lemma 1 uses $\dim T_1(A) = d$ in `Papers/0909.5347/main.tex:579--583`.

[3] The paper-facing general theorem is MPSTensor.iIndex_le_general_of_isPrimitivePaper in `TNLean/Wielandt/SourceTheorems/WielandtInequality.lean:394--473`. The formal equivalence between the primitive condition used by Sanz–Pérez-García–Wolf–Cirac and eventual full Kraus rank under normalization is in `TNLean/Wielandt/Primitivity/Equivalence.lean:169--191`.

These are the hypotheses in [SPGWC10, Theorem 1]. The older chosen-Kraus statements are still present, but only as corollaries obtained by taking $X = A_{i_0}$.[4]

3 Resolved one-step-subspace issue

The former distinction was small but genuine. Suppose first that the paper gives an invertible matrix $X \in S_1(A)$. Since X is a linear combination of the A_i, multiplication by X sends $S_n(A)$ into $S_{n+1}(A)$. Since X is invertible, this multiplication is injective on $M_D(\mathbb{C})$. This is the linear-algebra reason why the dimensions of the spaces $S_n(A)$ cannot stagnate for too long; it is the dimension-growth step used in the proof of the sharper bound.

The formal proof now uses this one-step-subspace version. It adjoins the chosen $X \in S_1(A)$ as a redundant generator and proves that the exact word spans and the Kraus rank are unchanged. The span equality is MPSTensor.wordSpan_oneStepAugment_eq, and the Kraus-rank equality is MPSTensor.krausRank_oneStepAugment, both in `TNLean/Wielandt/SpanGrowth/InvertibleWordSpan.lean`. This reduces the paper's hypothesis to the single-generator argument without changing the channel invariant being bounded.

The same issue appears in the non-invertible case. The source hypothesis gives a matrix $X \in S_1(A)$ which is singular but has a nonzero eigenvalue. The formal proof again uses the redundant-generator construction, so the theorem is stated for an arbitrary such X, not only for a distinguished Kraus matrix.

4 What is not a discrepancy

The replacement of d by $\mathrm{kr}(A) = \dim S_1(A)$ is not a paper error and not a formal weakening. It records explicitly the linearly independent Kraus count used by the source proof.

[4] The paper-facing declarations are MPSTensor.wordSpan_eq_top_of_isPrimitivePaper_of_mem_wordSpan_one_of_isUnit and MPSTensor.iIndex_le_of_mem_wordSpan_one_of_isUnit for the invertible case, and MPSTensor.wordSpan_eq_top_of_isPrimitivePaper_of_mem_wordSpan_one_of_noninvertible_eigenvector and MPSTensor.iIndex_le_sq_of_mem_wordSpan_one_of_noninvertible_eigenvector for the non-invertible case, all in `TNLean/Wielandt/SourceTheorems/WielandtInequality.lean`. The shorter chosen-generator statements in the same file are corollaries.

Nor is the use of the increasing spaces

$$T_n(A) = S_0(A) + S_1(A) + \cdots + S_n(A)$$

by itself a theorem-level deviation. Some intermediate formal results prove $T_{D^2}(A) = M_D(\mathbb{C})$ under normality. These results are weaker than the full quantum Wielandt theorem, but the paper-facing theorem proves the stated bound for $i(A)$, where $i(A)$ is defined using the spaces $S_n(A)$ themselves. [5]

Finally, the lower-level use of normality is not itself a mismatch. In these formal statements, normality is eventual full Kraus rank. Under the paper's normalization, the formal statements prove the equivalence between the paper's primitive condition and eventual full Kraus rank, so using normality as an intermediate hypothesis is legitimate.

5 Conclusion

The general quantum Wielandt bound has been formalized with the right index and with the paper's linearly independent Kraus count made explicit. The two improved estimates have also been upgraded to the paper's hypotheses: the distinguished matrix is any element of $S_1(A)$ with the required spectral property. The chosen-Kraus statements remain useful corollaries, but they should not be cited as the paper-facing theorem surface.

The remaining caution is theorem-surface hygiene. Cumulative-span declarations and bundled analysis chains are useful support lemmas for the proof, while the paper-facing results are the exact-length statements about $S_n(A)$ and the index $i(A)$.

References

[SPGWC10] M. Sanz, D. Pérez-García, M. M. Wolf, and J. I. Cirac. A quantum version of Wielandt's inequality. *IEEE Transactions on Information Theory*, 56(9):4668–4673, 2010.

[5] The cumulative-span theorem is `MPSTensor.cumulativeSpan_eq_top` in `TNLean/Wielandt/SpanGrowth/NonzeroTraceProduct.lean:164--197`; the paper-facing theorem for the spaces $S_n(A)$ is the general theorem cited above. The blueprint makes this distinction in `blueprint/src/chapter/ch07_wielandt.tex:461--490` and states the general bound in `blueprint/src/chapter/ch07_wielandt.tex:492--548`.